

Prospective source model of the 2026 M_w 7.3 Mexico (Chiapas) earthquake

A blind test of our MTUQ+WISP pipeline, produced before the USGS finite fault

Automated pipeline (17.Venezuela_2026)

Frozen 2026-07-18, hours after origin — for later comparison with the USGS solution

Purpose

The USGS had not yet posted a finite-fault model for `us7000t1bu` when this was produced. This is a prospective (blind) test: we run our teleseismic MTUQ moment-tensor + WISP finite-fault pipeline on open GSN data and freeze the prediction here so it can be compared with the USGS solution once posted. Everything below uses only teleseismic body + surface waves and the USGS W-phase moment tensor as the seed; no InSAR, regional, or geodetic data (which USGS will likely add, and which — per our Al Haouz study — tends to yield a smaller, more compact source).

1 Event and USGS preliminary benchmark (frozen)

Origin	2026-07-17T14:48:39.77 UTC, 14.604°N, 92.952°W, depth 18.6 km		
Magnitude	M_w 7.3 (W-phase); $M_0 = 1.1 \times 10^{20}$ N m; centroid depth 15.5 km		
Tectonics	Middle America subduction zone (Cocos plate), shallow thrust		
W-phase NP1 (steep)	strike 122.75, dip 63.86, rake 97.74		
W-phase NP2 (shallow)	strike 285.6, dip 27.19, rake 74.65 (likely megathrust interface)		

This is analogous in size to the 2024/2026 Atacama and 2026 Venezuela events we have modelled, but it is a shallow subduction thrust, so we apply the shallow-source lessons from our Al Haouz (Morocco) study.

2 Data

23 GSN broadband stations (networks IU, II, G, GE, ...) at 31–86°, fetched from IRIS. WISP uses 15 with usable P; MTUQ uses 11 three-component stations. The azimuthal coverage has a $\sim 87^\circ$ gap over the Pacific (few island stations) — adequate but not ideal.

M7.3 Mexico teleseismic stations (azimuthal-equidistant, rings 30/60/90 deg)

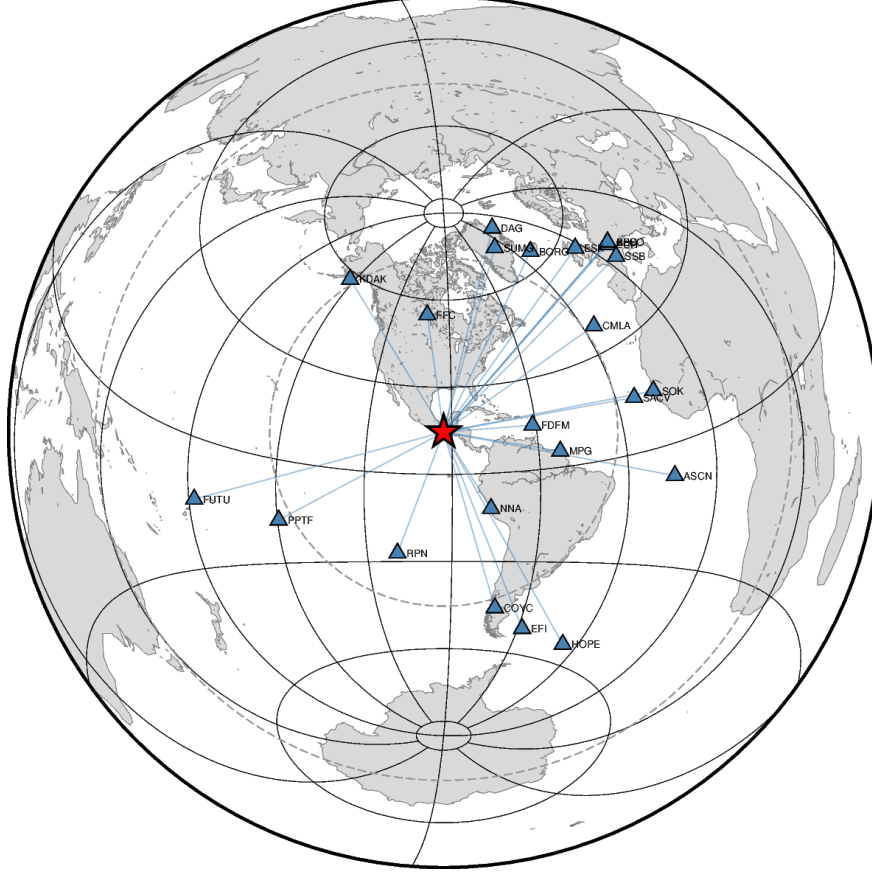


Figure 1: Teleseismic station distribution (azimuthal-equidistant, rings at 30/60/90°).

3 Moment tensor (MTUQ) — the shallow-thrust trap, a third time

For this shallow source the same pattern we documented on Al Haouz reappears, now at M_w 7.3:

MTUQ solution	strike/dip/rake	M_w	verdict
DC, body-only (adopted)	284/30/79	7.30	$\approx$ W-phase NP2 (shallow interface)
DC, surface waves	121/85/88	7.45	dip 85° (too steep — shallow indeterminacy)
full moment tensor	126/81/81	7.60	destabilised ($w = 0.90$, huge non-DC)
W-phase (USGS)	122.75/63.86/98 / 285.6/27.19/75	7.3	reference

The body-only DC recovers the W-phase mechanism and moment (284/30 matches the shallow NP2; M_w 7.30 matches M_{ww}). The surface-wave DC pushes the dip to 85° (the Kanamori–Given shallow dip-slip indeterminacy), and the full moment tensor is destabilised into a large spurious non-double-couple with M_w 7.60. This is the third independent confirmation of our rule: for a shallow source, invert body waves for the mechanism. We adopt the body-only DC.

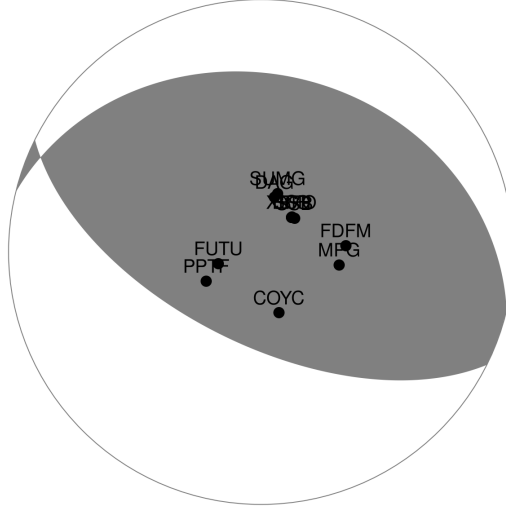


Figure 2: Adopted body-only DC moment tensor (284/30/79) with the station distribution — a shallow subduction thrust, consistent with the W-phase.

4 Finite fault (WISP), both nodal planes

WISP `auto_model` on both W-phase nodal planes, seeded from the W-phase moment (the key lesson from our Al Haouz ablation: seed from the authoritative M_{ww}). Teleseismic body + surface waves. We then applied the cross-correlation time-shift step (`shift_match`; not run by `auto_model` — a separate manual step) and re-inverted, which cut the misfit 13–16% on both planes with M_w unchanged (as on Al Haouz). Values below are the shifted models.

plane	strike/dip	M_w	misfit (pre→post shift)	$L \times W$ / peak slip	slip depth / T_{5-95}
NP1 steep	123/64	7.29	0.1120 → 0.0940	82×41 km / 3.43 m	1–34 km / 19.2 s
NP2 shallow (interface)	286/27	7.29	0.1114 → 0.0969	82×68 km / 2.54 m	3–24 km / 20.3 s

Table 1: Both planes fit almost identically — teleseismic waveforms are a weak plane discriminant, as we found for Al Haouz and Chile. For a subduction thrust the shallow-dipping NP2 is the megathrust interface and fits marginally better; we prefer NP2 but show both. Both give M_w 7.29 (matching M_{ww}), a compact updip asperity, and ~ 20 s duration. The time-shift step improved each fit ~ 13 –16% and slightly sharpened the slip (NP2 peak 2.29→2.54 m).

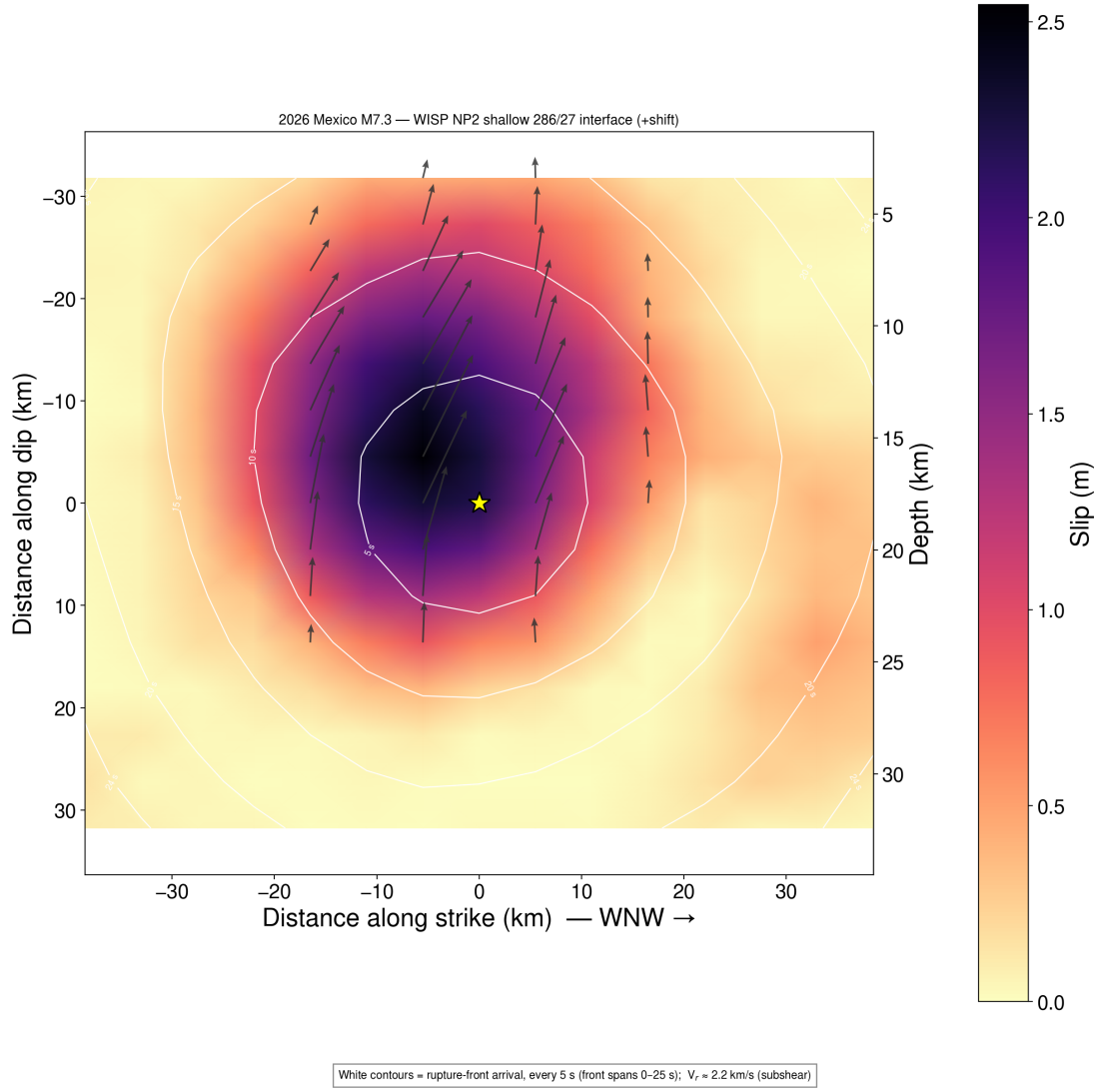


Figure 3: Preferred (NP2, shallow interface) on-fault slip at true 1:1 scale; star = hypocentre, white contours = rupture front every 5 s, arrows = thrust slip. Compact updip asperity, peak 2.5 m, slip ~ 5 –24 km depth (shifted model).

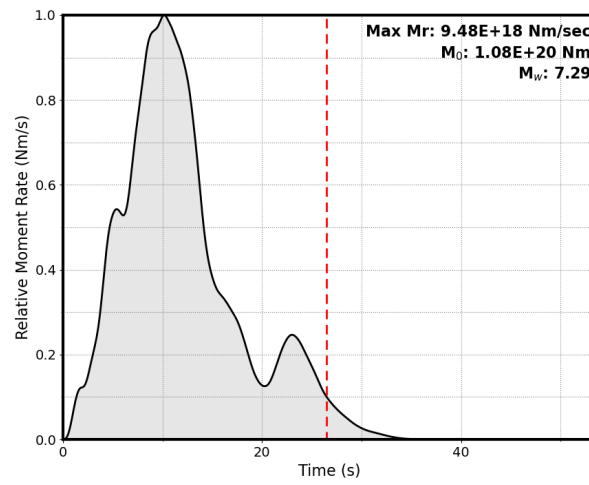


Figure 4: Moment-rate function (NP2): peak at ~ 11 s, total duration ~ 20 s.

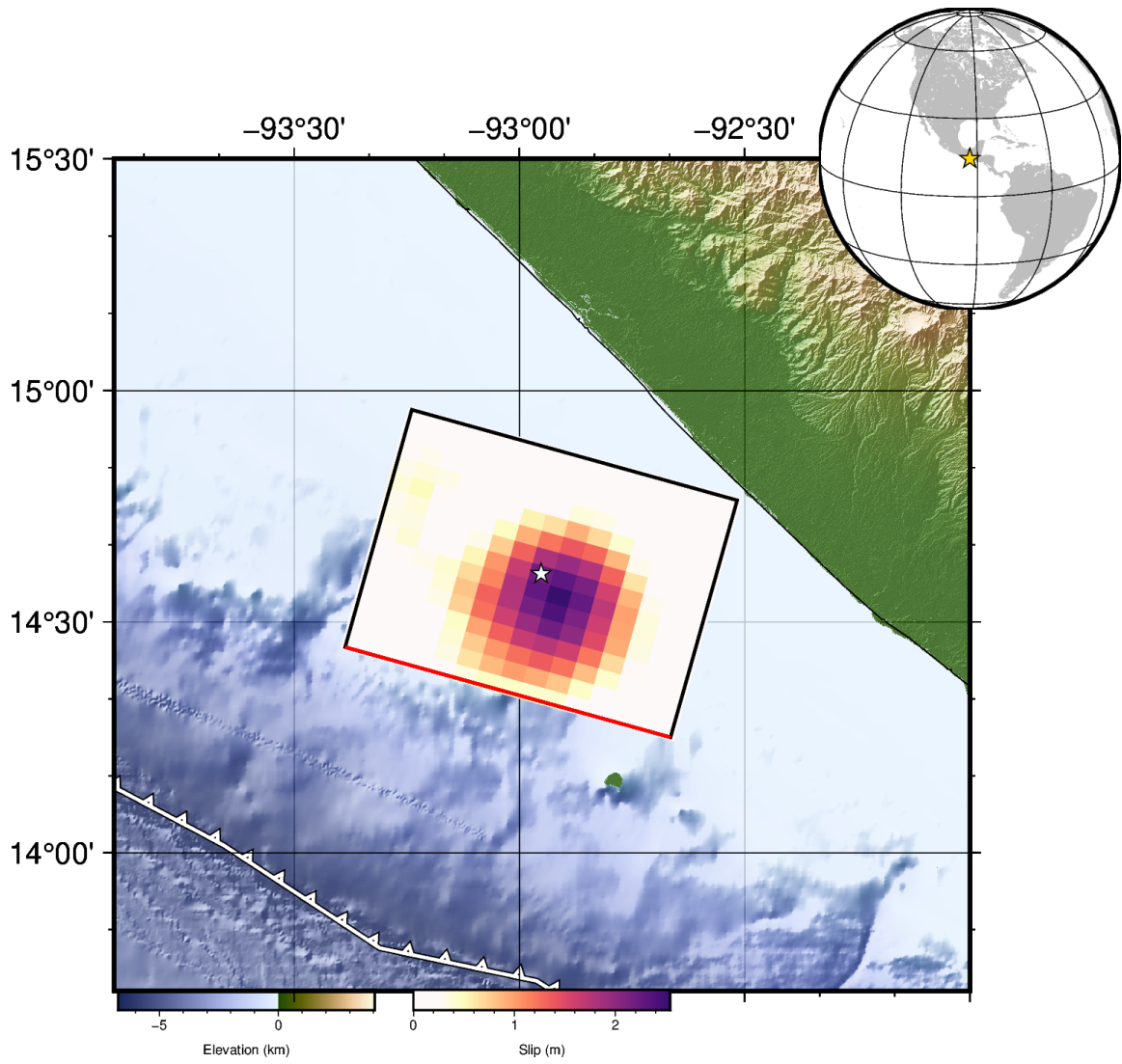


Figure 5: Map view of the NP2 slip projected to the surface.

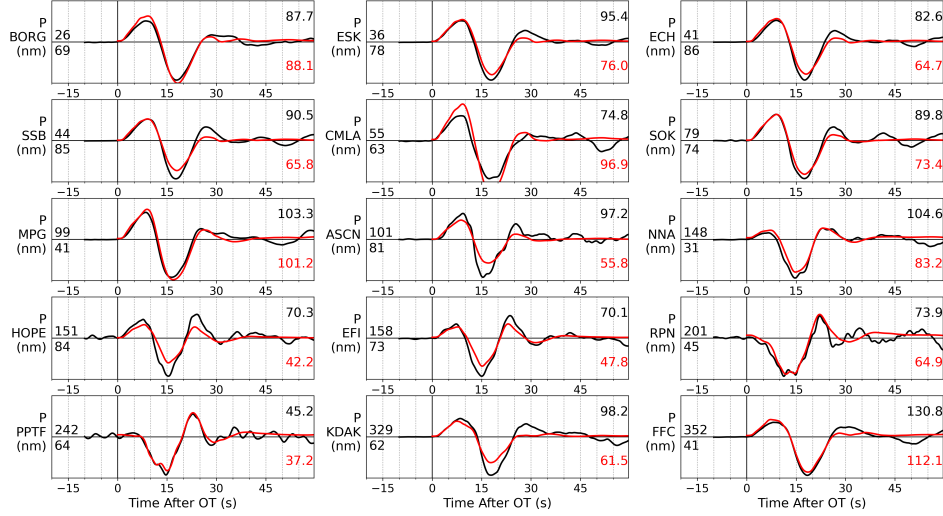


Figure 6: WISP teleseismic P-wave fits (NP2). Black observed, red synthetic.

5 Prospective prediction (compare against the USGS FFM when posted)

- Mechanism: shallow subduction thrust; preferred plane $\approx 285/27/75$ (megathrust interface), or steep conjugate $123/64$.
- M_w : 7.29–7.30 ($M_0 \approx 1.1 \times 10^{20}$ N m), matching M_{ww} .
- Peak slip: ~ 2.5 – 3.4 m; slip depth ~ 5 – 25 km; single compact updip asperity near/above the hypocentre.
- Duration: ~ 20 s, moment-rate peak at ~ 10 s.

Honest caveats. (1) Teleseismic-only: no InSAR/regional yet. Per our Al Haouz study, adding geodesy usually shrinks the source and raises peak slip. (2) Location is the catalogue hypocentre (not relocated to a geodetic centroid). (3) The $\sim 87^\circ$ Pacific azimuthal gap limits resolution on the down-dip/along-strike updip direction.

6 Comparison with the USGS finite fault (posted after our freeze)

USGS (D. Goldberg) posted their finite fault on 2026-07-17. Crucially it is also teleseismic-only (BODY 63, SURF 52, InSAR 0) — a true like-for-like comparison.

	USGS (Goldberg)	Ours (NP2_shift)	assessment
Fault plane	286/27/75	286/27/75	exact match
M_w	7.29	7.29	exact
Peak slip	2.85 m	2.54 m	close (−11%)
Slip depth (>15%)	4–25 km	3–24 km	excellent match
Moment-rate peak	10.2 s	10.2 s	exact
Data used	teleseismic-only	teleseismic-only	match
Fault template	125×65 km	82×68 km	USGS larger
Rupture velocity	1.85 km/s	2.08 km/s	ours faster
Duration T_{5-95}	35 s	20 s	USGS longer

Table 2: Blind-test outcome. The source fundamentals agree closely: the same nodal plane (we correctly preferred the shallow megathrust interface), the same M_w , the same slip depth, the same peak-slip magnitude, and the same main-pulse timing — both teleseismic-only. The differences are secondary modelling choices: USGS used a larger fault template (125 km) with a slower rupture velocity, producing a longer, more spread-out rupture; ours is more compact.

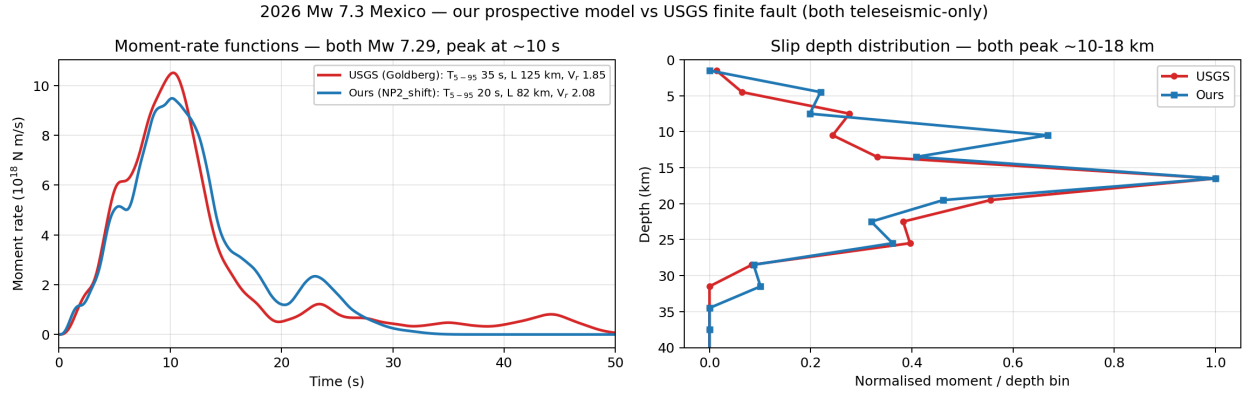


Figure 7: Left: moment-rate functions — the main pulse (which carries most of the moment) is nearly identical in timing and amplitude; USGS spreads the last ~15% of moment over a longer tail (bigger fault + slower V_r). Right: slip-vs-depth — the distributions nearly overlap, both peaking at ~16 km.

Where our forecast missed (honest). Our two caveats predicted the opposite of what USGS did: we expected them to add InSAR and produce a more compact source. Instead their model is teleseismic-only and longer than ours. So our prediction was, if anything, conservatively compact — the mechanism, moment, slip depth and main-pulse timing all validated, while the total rupture length/duration is a genuine model-choice difference (larger template + slower V_r) on which the teleseismic data are weakly constraining.